Customer Retention Enhancement through Predictive Analysis

Author: Firdausi Jibrin **Date:** 01 March, 2026 **Organization:** Lloyds Banking Group (Forage Project)

Introduction

Customer churn occurs when customers stop doing business with a company, leading to potential revenue loss. Identifying and preventing churn is crucial for maintaining a stable customer base.

Aim and Objectives

This project aims to:

- Analyze customer data
- Identify patterns associated with churn
- Prepare the dataset for predictive modeling.

Dataset Summary and Rationale

The dataset contains customer information, including demographic data, account details, and churn status.

Key variables:

- 'Customer ID': Unique identifier for each customer
- 'Age', 'Gender': Demographic features
- 'TransactionID', 'AmountSpent', 'ProductCategory': Financial Information
- 'ChurnStatus': Target variable indicating if the customer left (1) or stayed (0)

Rationale for inclusion

- These variables have been shown to influence customer behaviour and churn likelihood.
- Financial and account activity features help identify patterns that predict churn.
- Demographic variables allow segmentation and targeted retention strategies.

Initial Data Inspection

Before cleaning, we inspect the dataset to understand distributions, identify missing values, and detect patterns.

In [269…
```python
import pandas as pd
```

In [270…
```python
# Load dataset
df = pd.ExcelFile("Customer_Churn_Data_Large.xlsx")
# Load all sheet names in the Excel file
df.sheet_names
```

Out[270…
```
['Customer_Demographics',
 'Transaction_History',
 'Customer_Service',
 'Online_Activity',
 'Churn_Status']
```

In [271…
```python
# Load specific sheets
df_demo = pd.read_excel(df, sheet_name='Customer_Demographics')
df_trans = pd.read_excel(df, sheet_name='Transaction_History')
df_service = pd.read_excel(df, sheet_name='Customer_Service')
df_online = pd.read_excel(df, sheet_name='Online_Activity')
df_churn = pd.read_excel(df, sheet_name='Churn_Status')
```

In [272…
```python
print(df_demo.columns)
print(df_trans.columns)
print(df_service.columns)
print(df_online.columns)
print(df_churn.columns)
```

```
Index(['CustomerID', 'Age', 'Gender', 'MaritalStatus', 'IncomeLevel'], dtype='obj
ect')
Index(['CustomerID', 'TransactionID', 'TransactionDate', 'AmountSpent',
       'ProductCategory'],
      dtype='object')
Index(['CustomerID', 'InteractionID', 'InteractionDate', 'InteractionType',
       'ResolutionStatus'],
      dtype='object')
Index(['CustomerID', 'LastLoginDate', 'LoginFrequency', 'ServiceUsage'], dtype='o
bject')
Index(['CustomerID', 'ChurnStatus'], dtype='object')
```

In [273…
```python
df = df_demo.copy()
```

In [274…
```python
# All datasets were merged using the CustomerID column to create a unified datas
df = pd.merge(df, df_trans, on='CustomerID', how='left')
df = pd.merge(df, df_service, on='CustomerID', how='left')
df = pd.merge(df, df_online, on='CustomerID', how='left')
df = pd.merge(df, df_churn, on='CustomerID', how='left')
```

In [275…
```python
# Check columns
df.columns
```

Out[275…
```
Index(['CustomerID', 'Age', 'Gender', 'MaritalStatus', 'IncomeLevel',
       'TransactionID', 'TransactionDate', 'AmountSpent', 'ProductCategory',
       'InteractionID', 'InteractionDate', 'InteractionType',
       'ResolutionStatus', 'LastLoginDate', 'LoginFrequency', 'ServiceUsage',
       'ChurnStatus'],
      dtype='object')
```

In [276…
```python
# View Dataset
df.head()
```

Out[276…

	CustomerID	Age	Gender	MaritalStatus	IncomeLevel	TransactionID	TransactionDa
0	1	62	M	Single	Low	7194	2022-03-2
1	2	65	M	Married	Low	7250	2022-08-(
2	2	65	M	Married	Low	9660	2022-07-2
3	2	65	M	Married	Low	2998	2022-01-2
4	2	65	M	Married	Low	1228	2022-07-2

In [277…

```
# Dataset Overview
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6812 entries, 0 to 6811
Data columns (total 17 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        6812 non-null   int64
 1   Age               6812 non-null   int64
 2   Gender            6812 non-null   object
 3   MaritalStatus     6812 non-null   object
 4   IncomeLevel       6812 non-null   object
 5   TransactionID     6812 non-null   int64
 6   TransactionDate   6812 non-null   datetime64[ns]
 7   AmountSpent       6812 non-null   float64
 8   ProductCategory   6812 non-null   object
 9   InteractionID     5204 non-null   float64
 10  InteractionDate   5204 non-null   datetime64[ns]
 11  InteractionType   5204 non-null   object
 12  ResolutionStatus  5204 non-null   object
 13  LastLoginDate     6812 non-null   datetime64[ns]
 14  LoginFrequency    6812 non-null   int64
 15  ServiceUsage      6812 non-null   object
 16  ChurnStatus       6812 non-null   int64
dtypes: datetime64[ns](3), float64(2), int64(5), object(7)
memory usage: 904.8+ KB
```

In [278…

```
# Summary Statistics
df.describe()
```

Out[278…

	CustomerID	Age	TransactionID	TransactionDate	AmountSpent	Inter
count	6812.000000	6812.000000	6812.000000	6812	6812.000000	520
mean	500.169260	43.274516	5497.323253	2022-07-01 02:11:29.136817408	251.620527	592
min	1.000000	18.000000	1000.000000	2022-01-01 00:00:00	5.180000	201
25%	247.000000	30.000000	3223.500000	2022-04-02 00:00:00	127.100000	387
50%	505.500000	44.000000	5515.000000	2022-07-01 00:00:00	251.845000	590
75%	750.000000	56.000000	7675.250000	2022-09-30 00:00:00	375.280000	785
max	1000.000000	69.000000	9997.000000	2022-12-31 00:00:00	499.860000	999
std	286.704642	15.286788	2584.768541	NaN	142.901693	233

In [279…

```
# Missing values and duplicates
df.isnull().sum()
df.duplicated().sum()
```

Out[279…

```
np.int64(0)
```

Exploratory Data Analysis

EDA was conducted to understand the dataset and identify patterns related to customer churn.

In [280…

```
import matplotlib.pyplot as plt
import seaborn as sns
```

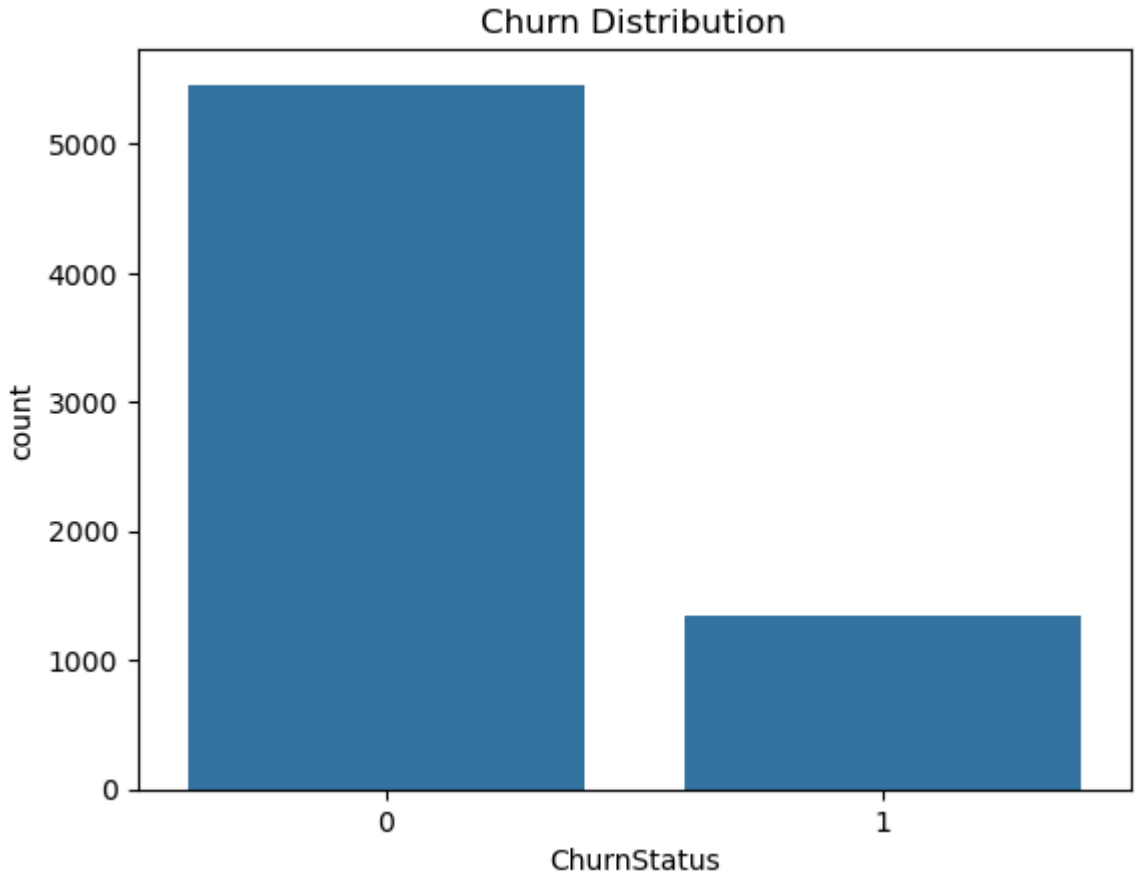
Demographics Visualizations

In [281…

```
# Churn Distribution
sns.countplot(x='ChurnStatus', data=df)
plt.title('Churn Distribution')
```

Out[281…

```
Text(0.5, 1.0, 'Churn Distribution')
```

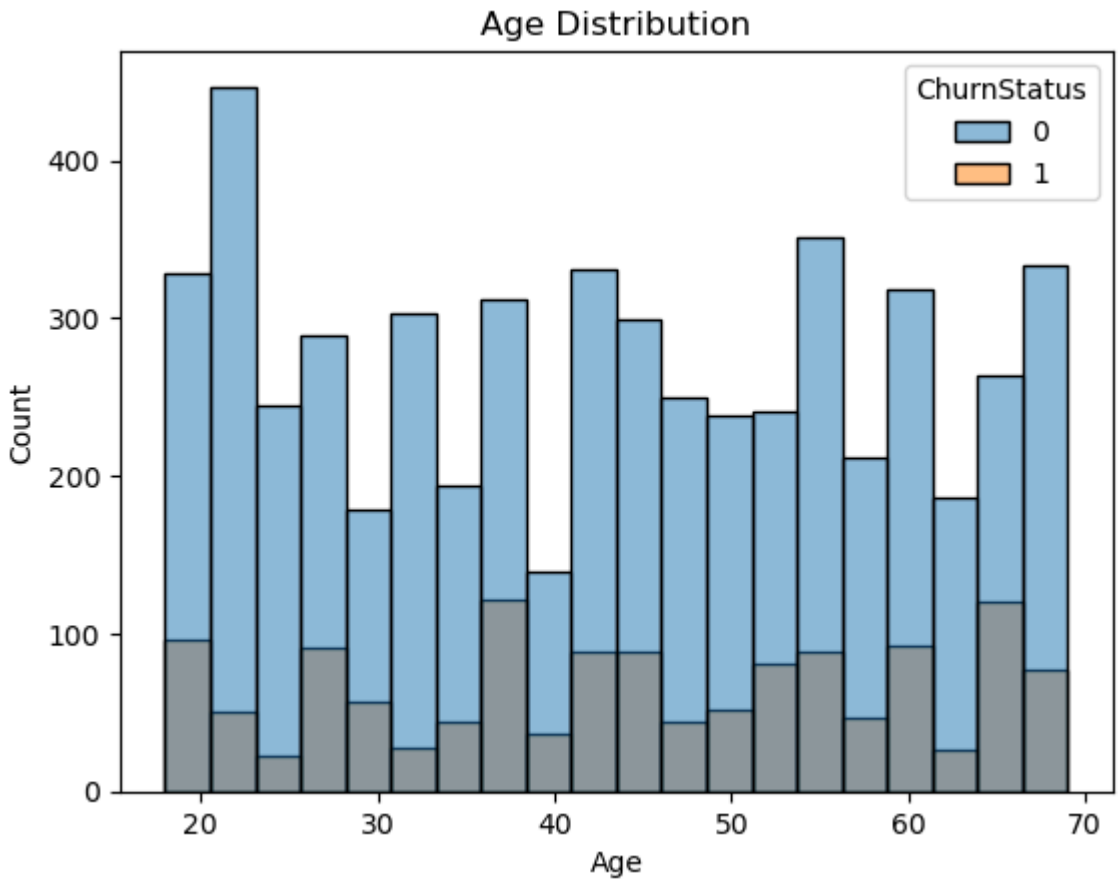


In [282…

```
# Age Distribution
sns.histplot(data=df, x='Age', hue='ChurnStatus', bins=20)
plt.title('Age Distribution')
```

Out[282…

```
Text(0.5, 1.0, 'Age Distribution')
```

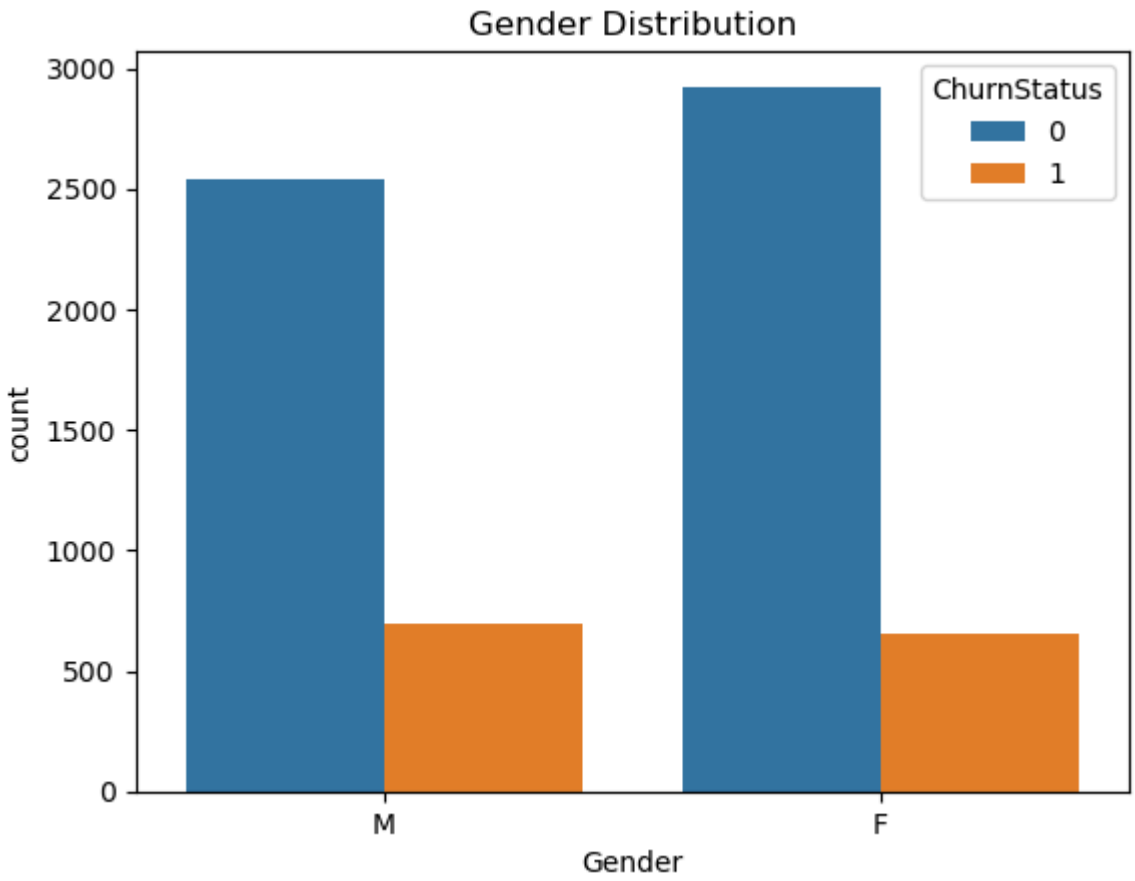


In [283…

```
# Gender Distribution
sns.countplot(x='Gender', hue='ChurnStatus', data=df)
plt.title("Gender Distribution")
```

Out[283…

```
Text(0.5, 1.0, 'Gender Distribution')
```

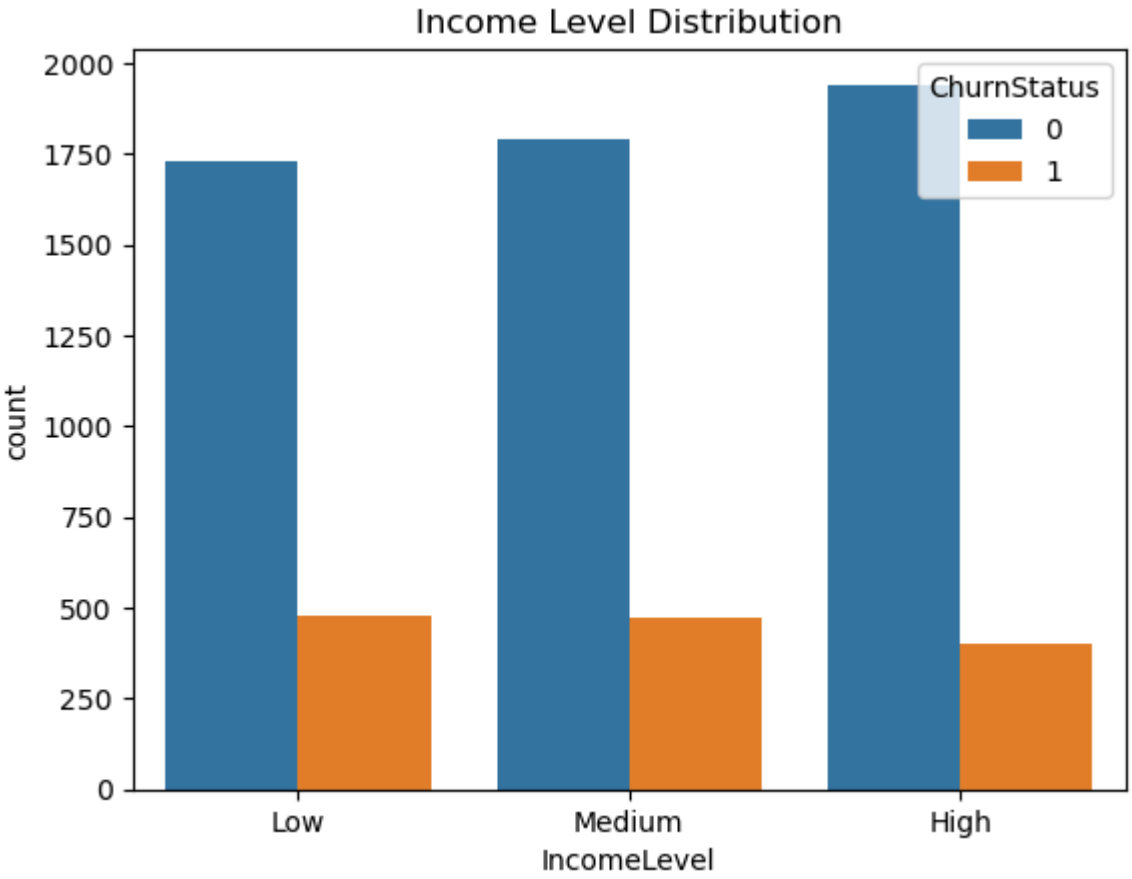


In [284…

```
# Income Level Distribution
sns.countplot(x='IncomeLevel', hue='ChurnStatus', data=df)
plt.title("Income Level Distribution")
```

Out[284…

```
Text(0.5, 1.0, 'Income Level Distribution')
```



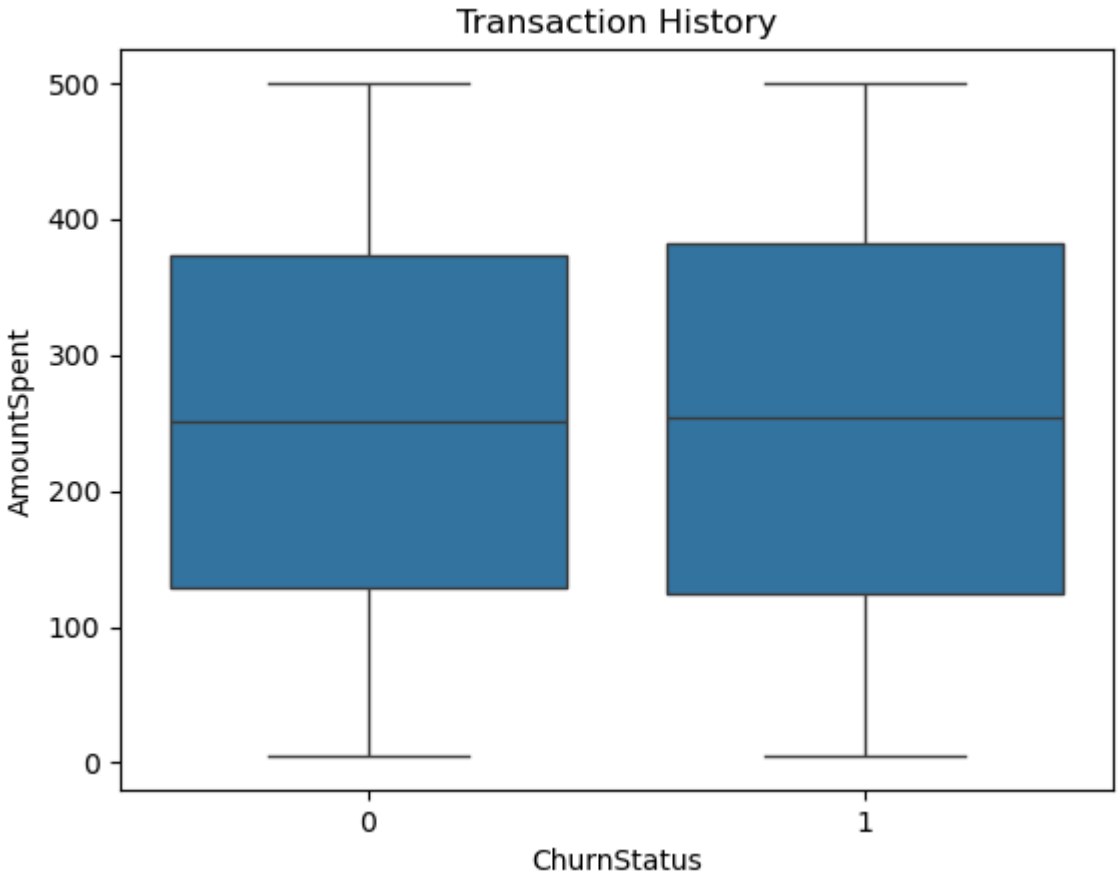
Transaction Visualizations

In [285…

```
# Transactions vs Churn
sns.boxplot(x='ChurnStatus', y='AmountSpent', data=df)
plt.title("Transaction History")
```

Out[285…

```
Text(0.5, 1.0, 'Transaction History')
```



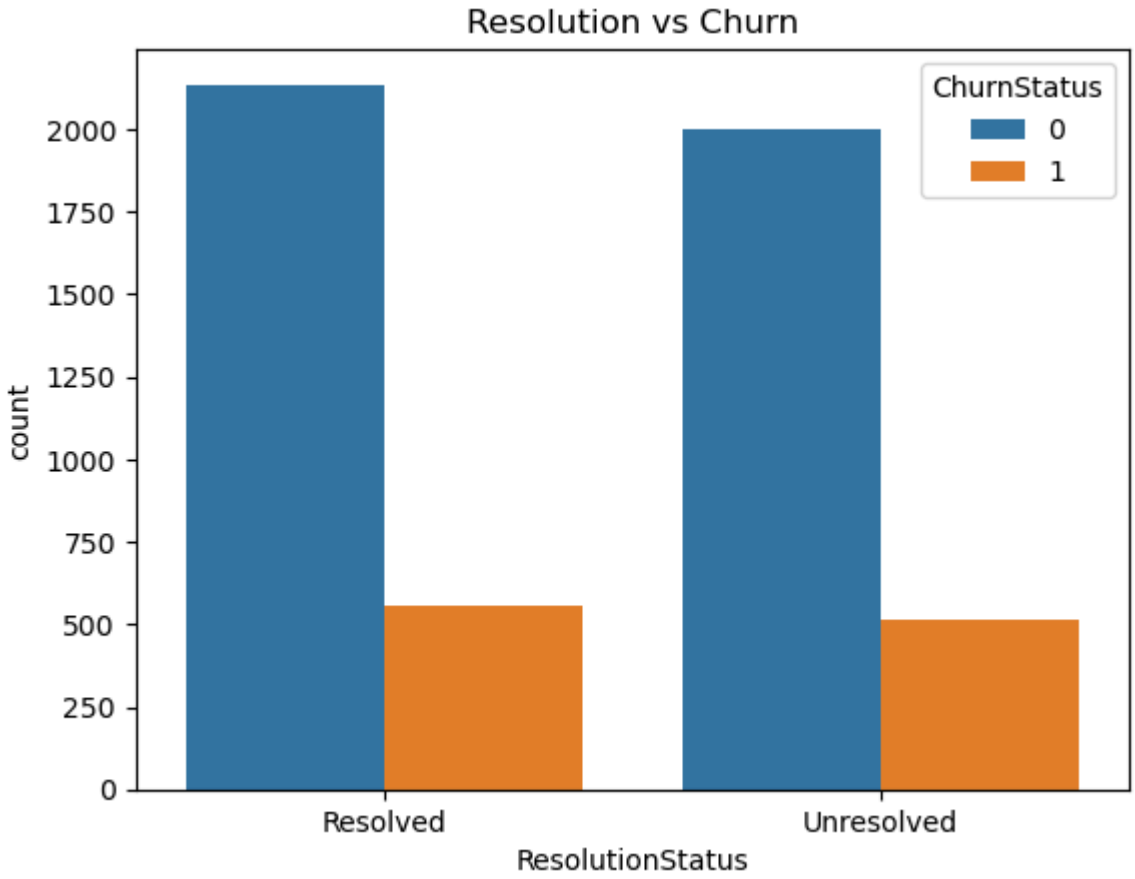
Customer Service Visualizations

In [286…

```
# Resolution Status
sns.countplot(x='ResolutionStatus', hue='ChurnStatus', data=df)
plt.title("Resolution vs Churn")
```

Out[286…

```
Text(0.5, 1.0, 'Resolution vs Churn')
```



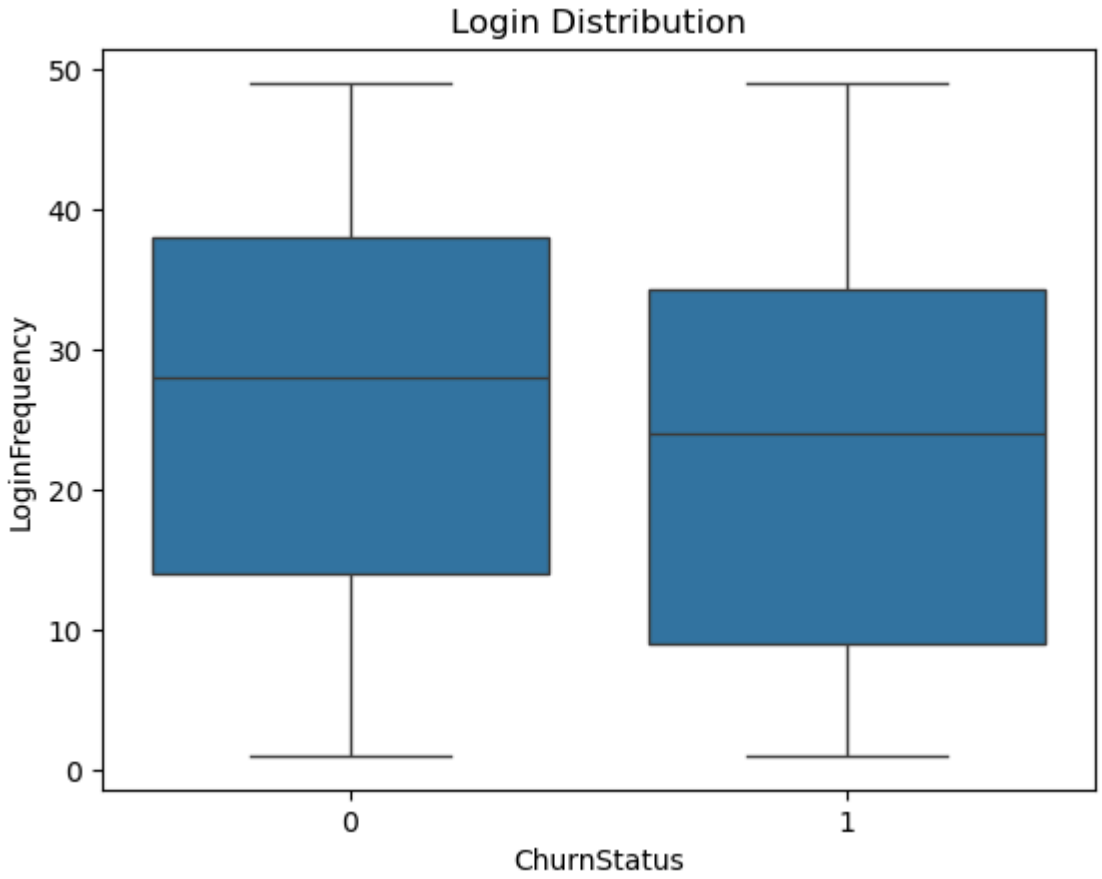
Online Activity Visualizations

In [287…

```
# Login Distribution
sns.boxplot(x='ChurnStatus', y='LoginFrequency', data=df)
plt.title("Login Distribution")
```

Out[287…

```
Text(0.5, 1.0, 'Login Distribution')
```

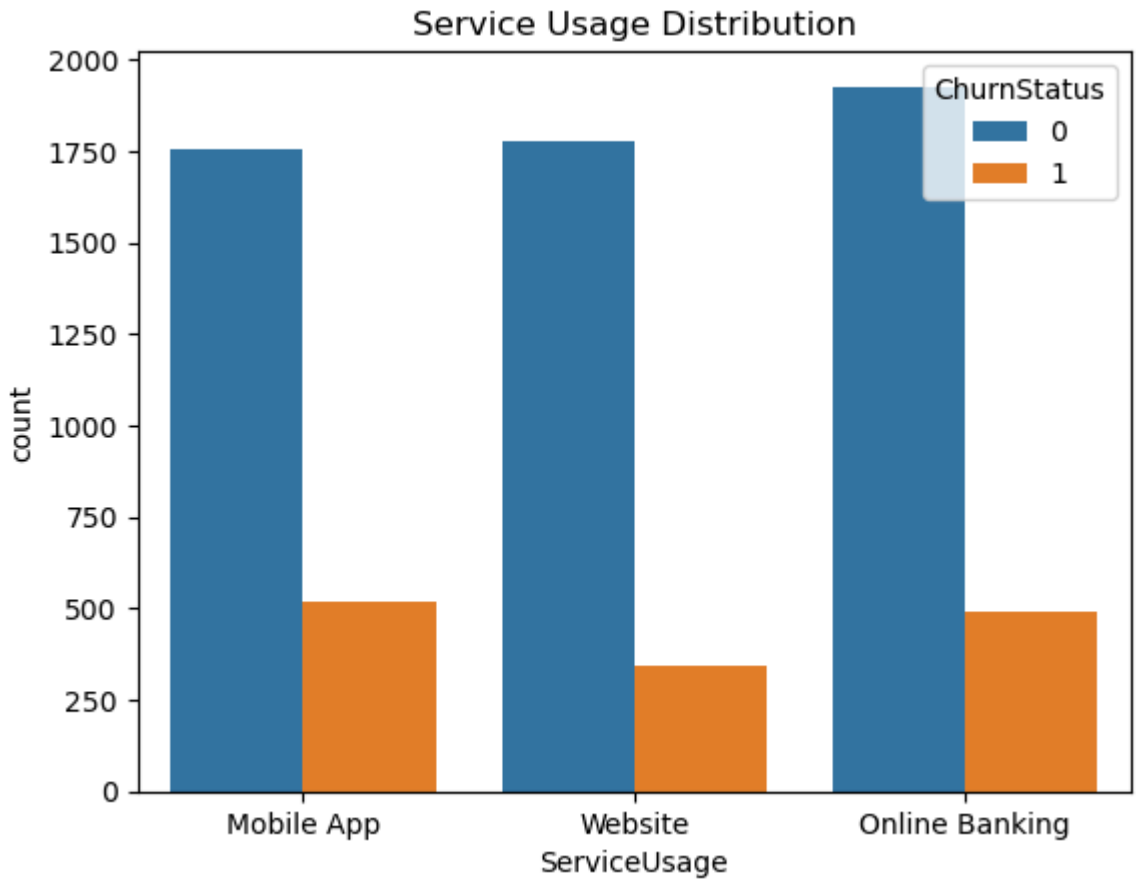


In [288…

```
# Service Usage Distribution
sns.countplot(hue='ChurnStatus', x='ServiceUsage', data=df)
plt.title("Service Usage Distribution")
```

Out[288…

```
Text(0.5, 1.0, 'Service Usage Distribution')
```



Correlation Heatmap

This helps identify which features relate to churn.

In [ ]:
```
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True)
plt.title("Correlation Heatmap")
```

Out[ ]:
```
Text(0.5, 1.0, 'Correlation Heatmap')
```

Key Findings

- According to the Churn Distribution Visualization, the number of customers who stayed is significantly higher than the number of customers who churned or left
- The Age Distribution visualization also shows the number of customers who stayed encompasses those who age. customers who churned where within the range of 100 in number compared to those who staywed.
- The Gender Distribution visualization showed the female gender of those who stayed were significantly higher than the male gender.
- In relation to Income Level of customers, those with higher income level were greater than medium income level, and medium income earners were slightly higher than low income earners. High income earners stayed most.
- In view of the amount spent by customers, customers who left spent slightly higher than those who stayed.
- According to the resolution status of the data, the number of resolved cases for those who stayed was slightly higher than those who left. The visualization shows those who churned having more unresolved cases than those who stayed.
- The Login distribution shows those who stayed had more login distribution than those who churned.
- The service usage distribution showes customers who used online banking where more in numbers than customers who used website or mobile app. Customers who churned mostly made use of the website service.

Data Cleaning and Preprocessing

The following preprocessing steps were performed:

In [ ]:
```
# Duplicate Removal
df = df.drop_duplicates(inplace=True)
```

In [ ]:
```
# Encoding
categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
categorical_cols = [col for col in categorical_cols if col not in ['CustomerID',

df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
```

In [ ]:
```
# Scale numeric columns
from sklearn.preprocessing import StandardScaler
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns.tolist()
scaler = StandardScaler()
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])
```

In [268…
```
df.to_csv("Cleaned_Churned_Dataset.csv", index=False)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[268], line 1
----> 1 df.to_csv("Cleaned_Churned_Dataset.csv", index=False)

AttributeError: 'NoneType' object has no attribute 'to_csv'
```